Higher-arity Koblitz extraction versus matched rho

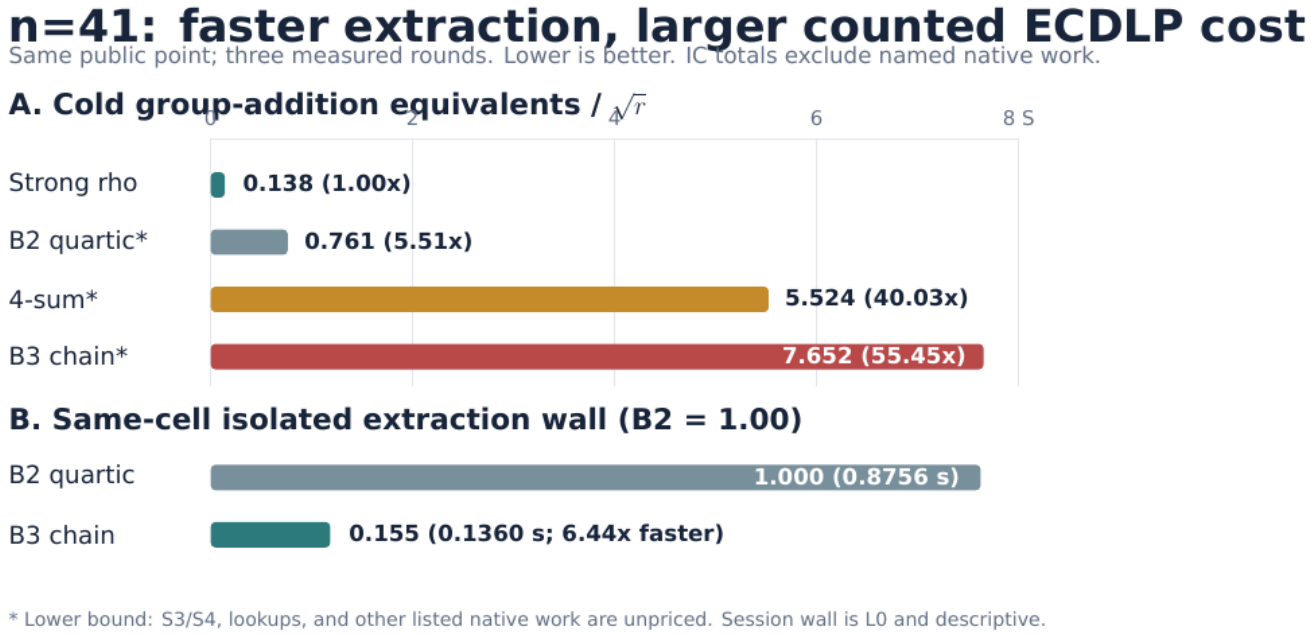
Audited n=41 discriminator for the ECC2K-130 scaling blocker

crypto research evidence handoff

2026-10-13

Decision

The two-S3-chain kernel is correct and makes the stride-ordered extraction primitive **6.44x faster** than the quartic kernel in a same-cell isolated probe. It does **not** improve total algorithmic cost: the end-to-end candidate is **55.447x matched strong rho** in counted group-addition equivalents before pricing its S3 and lookup work.



Controlled change

For every stride-ordered pair cell `(s1,s2)`, B3 replaces one quartic S4 root solve with the union of two S3 root solves through `R+s1` and `R−s1`. The factor base, pair-cell order, targets, relation verifier, linear algebra and scalar replay are held fixed. Release tests establish split-quartic root-set identity and verified recovery on planted, random and negative cases.

Evidence

Both isolated arms reached the same verified relation after 78,041 cells.

kernel	root work	point adds	extraction wall
B2 quartic	78,041 S4	uncharged	0.8756 s
B3 chain	156,082 S3	234,129	0.1360 s

The ecbench session ran one public target, one warmup, and three paired measured rounds. All 20 executions verified; all 15 measured deterministic runs replayed identically.

arm	mean S	interval	ratio to rho
strong rho	0.1380	[0.088, 0.202]	1.000
B2 quartic	0.7609	[0.662, 0.888]	5.513
4-sum	5.5241	[4.327, 7.190]	40.027
B3 chain	7.6521	[6.616, 8.989]	55.447

The rho mean is 0.997x the signed-Frobenius theory floor. B2 and B3 generated the same 58 relations with identical trial and rank counts. Across measured runs both visited 5,109,566 pair cells; B3 therefore added 15,328,698 charged group operations. B3/B2 is 10.057x in counted work even though descriptive pipeline wall is 0.150x.

Blocker and next evidence gate

The remaining blocker is pair-cell complexity, not quartic implementation. With about 88,096 cells per accepted relation here, materializing three group adds per unchanged cell exceeds rho before native S3 and lookup work is priced. A credible next candidate must reduce cells or amortize point construction across cells/targets.

The n=83 gate is unavailable: ecbench reports that no factor-base plug-in runs on a wide field (`n > 62`), and only strong rho and the claw method currently have wide dispatch. Therefore these results establish neither an n=83 scaling exponent nor an ECC2K-130 transfer claim.

Qualification

Counts are primary. The ecbench run is L0 because PSI telemetry is unavailable; wall ratios are descriptive. IC totals are lower bounds because S3/S4 and lookup work is unpriced. This is one n=41 rung on one host, not independent replication. The complete session, receipt, protocol, comparisons and machine- readable summary accompany this PDF.